

A Comparative Study for Breast Cancer Prediction using Machine Learning and Feature Selection

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feature-selection

Abstract—While there are many factors which could contribute to the occurrence of breast cancer, it is very difficult to attribute the exact environmental and other factors contributing to it, but still it has significance in determining the occurrence of cancer. Using machine learning techniques and regular diagnosis information, we can achieve our goal of assessing the risk of occurrence of breast cancer. Cancer data sets contain many attributes of patient information, but not every feature is relevant in predicting cancer. Feature selection techniques are useful in such scenarios for retaining the relevant feature set. In this paper we are doing a comparative study of the effect of feature selection techniques on the accuracies given by existing machine learning algorithms. For this purpose we have considered the following machine learning algorithms - Logistic Regression, Naive Bayes and Random Forest. The following feature selection techniques have been considered - Sequential Forward Feature Selection, Recursive Feature Elimination, f-test and correlation. The publicly available Breast Cancer Wisconsin (Diagnostic) Data Sets from UCI Repository have been used in this work. The results show that random forest algorithm gives the highest accuracy with feature selection. Furthermore f-test gives better results for the smaller dataset and Sequential Forward Selection for the larger dataset.

Index Terms—breast cancer prediction, feature selection, machine learning

I. INTRODUCTION

Breast cancer is one of the leading causes of women mortality world wide. Though cervical cancer was the most common cancer among Indian women, the incidence of breast cancer has surpassed cervical cancer as the leading cause of death in India. It is common among the younger age groups (in their thirties and forties) and mostly occurs in women aged from 30 to 69. There are many features to be taken into consideration while validating breast cancer which include biopsy, family history and list goes on. The risk of occurrence of breast cancer may change with respect to change in any of these features. A regular diagnosis helps us to accumulate the variations and it's impact on a particular patient (patient history of health). All these together would lead to a conclusion on the status of the patient's health thereby giving us better results in accurately predicting the risk factor of the instance(patient). In this paper, we have implemented Logistic Regression, Naive Bayes and Random forest algorithms as base classifiers for predicting

the occurrence of breast cancer in a particular patient. After obtaining the accuracies for the above mentioned classifiers, we use feature selection algorithms like Sequential Feature Forward Selection, Recursive Feature Elimination, f-test and correlation methods and obtain the accuracies for each of the base classifiers on the reduced feature set given by the above mentioned feature selection algorithms. We then do a comparative study to obtain the combination of the above mentioned machine learning algorithms and feature selection techniques which gives us the best performance. The datasets used in this paper are Breast Cancer Wisconsin (Diagnostic) Data Sets from UCI Repository.

II. LITERATURE SURVEY

Many models have been proposed in the literature which use different feature sets and machine learning models for the prediction of breast cancer. The Gail model[1] uses non-genetic risk factors including: age, hormonal history (age at menarche, age at the time of birth of first child, number of past breast biopsies, number of breast biopsies showing atypical hyperplasia, race/ethnicity, Family history (limited) as the features for predicting breast cancer. The algorithm used in this model for classification is logistic regression. The drawback of this model is that it does not consider genetic risk factors which play an important role in the occurrence of breast cancer. The BRCAPRO Model[2] predicts the patients lifetime risks of developing breast and ovarian cancer in the next 5 years based on the probability that they carry a BRCA1 or BRCA2 gene mutation. This model also considers ethnicity for each family member, and both the maternal and paternal family history. The cancer probabilities are calculated based on the probability that the family has a mutation in either of the two genes. This model uses Bayes Theorem for calculation. The Tyrer-Cuzick model [3] predicts both the likelihood that the patient is having a breast cancer susceptible gene through heredity and the probability of carrying the aforementioned gene mutation. Family history, Hormonal history, Benign breast disease, Body mass index, Genetic factors, Age are some of the factors considered in this model. This model estimates the risk using Bayes Theorem. Feature selection algorithms play an important role in extracting features which are important for improving the performance of a model. The effect of feature selection algorithms has been studied in the

scenario of cancer prediction as well as in other domains. In one of the works, a method call Kernel F-test Feature Selection algorithm [4] has been proposed for Breast cancer prediction. Kernel functions have been used in this method to increase the dimensions of the input features for clear separation of non-linearity. For all the features F-score values in the kernel space are calculated. The mean kernel F-score value is used as the threshold. SVM classifier is used for classification in this model. A study conducted in the domain of educational data mining has identified various combinations of machine learning algorithms and feature selection techniques [5] which perform well in the scenario of educational data mining. Some of the combinations include Decision trees & Correlation based feature selection, Voted perceptron & One Rule, Random Forest Classification & Gain Ratio. Another study[6] shows the impact of feature selections on traditional machine learning classifiers in making predictions as a result of which performance is improved by decreasing the computational complexity and accuracy. It considers filter, wrapper and hybrid feature selection methods and it's affect on machine learning algorithms like Decision trees, Bayesian Networks and SVM. A comparative analysis has been done in one another paper[7] to extract the best features out of a large dataset. It experiments with trial and error exhaustive search, Sequential forward and backward search wit Neural Networks and Combination of heuristics and SVM. It has been concluded that SVM with K-means, which reduces the feature space from 32 to 6, gives a good accuracy of 97.38 percent is better than other time consuming methods considered in the paper. In paper[8] a fusion of multi-classifiers and feature reduction transformation method on three publicly available datasets- WBC,WDBC,WPBC has been introduced. The results show that the superior fusion for WBC is MLP and J48 with PCA, for WDBC is Single classifiers(SMO) or combination of SMO and MLP or SMO and IBK and combination of MLP, J48, SMO and IBK is superior to the other classifiers in WPBC. The objective of this study[9] is to use feature selection to remove unrelated and redundant features for improving classifier performance. A threshold-based feature selection techniques has been proposed and the performance has been compared to classifiers on three datasets based on software metrics and defect .The models have been evaluated using eight performance metrics and it has been concluded that threshold-based feature selection using the AUC parameter provides generally good performance. Furthermore, it has been found that even after doing away with 96 percent of the available software metrics, the classification models give better results in most of the cases. Machine learning is also used in various other fields.In paper[10] they have used machine learning classifiers such as K-nearest neighbor, decision tree-C4.5, Naive Bayes, and the random forest using 10-fold cross validation method for early predictions of coronary heart events. In the paper [11] they have used MR images generated by CAD systems for Alzheimer's disease and have proposed a method based on multivariate approaches handling the whole image as a single observation where segmentation

of the gray matter region in the image is performed using Gaussian Mixture Models and extraction of the score vectors is done using Partial Least Squares. SVM classifier is used for the classification of the dataset and the proposed method is shown have better performance than the existing models. In this paper [12] they have used Lung cancer CT scan images to predict if the cancer is benign or malignant using a pre-trained convolutional neural network named AlexNet and have achieved an accuracy of 98 percent for the same. According to the paper [13], some particular features like calories, sugars, carbohydrates, vitamin A, vitamin C and price help in determination of quality of a particular brand of juice. This means that a set of features which highly influence the predictions should be taken for constructing the models.

III. PROPOSED METHOD

In this paper we propose to do a comparative analysis of the performance of various feature selection algorithms with different machine learning models. We have considered three machine learning algorithms-Random Forest, Naive Bayes and Logistic Regression. We compare the accuracies given by these models before and after applying feature selection algorithms, namely f-test, Recursive Feature Elimination (RFE), Sequential Feature Selection (SFS) and Correlation . Figure1 shows the flowchart for the proposed method.

A. Dataset

We have used publicly available Breast Cancer Wisconsin (Diagnostic) Data Sets from UCI Repository. It consists of the following datasets:

- Breast cancer Wisconsin dataset - This dataset consists of 699 instances with 11 attributes, namely Sample code number, Clump Thickness, Uniformity of Cell Size, Uniformity of Cell Shape, Marginal Adhesion, Single Epithelial Cell Size, Bare Nuclei, Bland Chromatin, Normal Nucleoli, Mitoses, Class. The target feature Class consists of 2 labels, 2 for benign and 4 for malignant.
- WDBC dataset - This dataset consists of 569 instances with 32 attributes, namely id, diagnosis, radius mean, texture mean, perimeter mean, area mean, smoothness mean, compactness mean, concavity mean, concave points mean, symmetry mean, fractal dimension mean, radius standard error, texture standard error, perimeter standard error, area standard error, smoothness standard error, compactness standard error, concavity standard error, concave points standard error, symmetry standard error, fractal dimension standard error, radius worst, texture worst, perimeter worst, area worst, smoothness worst, compactness worst, concavity worst, concave points worst, symmetry worst, fractal dimension worst. The target feature diagnosis consists of 2 labels, B for benign and M for malignant.

B. Feature Selection Techniques

- 1) **Sequential Feature Selection** Sequential feature selection algorithm is a greedy search algorithm that is

used to reduce an n-dimensional feature space to a m-dimensional feature subspace where $m < n$. It is a wrapper approach which automatically selects a subset of features that improves the predictive performance of the classifier. It excludes or includes features one at a time based on the classifier performance until a feature subset of the size m is obtained.

- 2) **Recursive Feature Elimination** It is a combination of filter and wrapper method and is based on the concept of recursively constructing a model and choosing a feature, keeping the feature aside and then repeating the process with the remaining set of features. This process is repeated until all the features in the dataset are considered. Based on when they were eliminated, the features are ranked. It is a greedy method for finding the best performing subset of features where it's stability depends considerably on the type of model used for ranking features at each iteration.
- 3) **F-Test** F-test calculates the ratio of variance between two groups and the within the group for a feature. The groups in this case are the instances with the same target value. Greater value of f-test means that the distances within the groups are less and distances between the groups are more. In f-test the features are ranked based on higher values of f-score, which is the ratio of variance between two groups and the within the group.

C. Machine Learning Algorithms

- 1) **Random Forest** It creates a group of decision trees from randomly selected subset of training set. It then combines the results from different decision trees to predict the target class of the test instance. It works well because a single decision tree may be susceptible to noise, but the collective outputs of multiple decision trees reduce the effect of noise giving better results. It helps to control over-fitting which in turn leads to better predictive accuracy.
- 2) **Naive Bayes** Naive Bayes is a supervised machine learning algorithm based on the concept of Bayes theorem. It assumes conditional independence between every pair of features given the value of the target variable. The following is the equation for Naive Bayes algorithm:

$$y = \underset{y}{\operatorname{argmax}} P(y) \prod_{i=1}^n P(x_i|y) \quad (1)$$

There are different Naive Bayes algorithms. We have considered Gaussian Naive Bayes in this paper. When the predictors are not discrete and takes up a continuous value, we assume that they are sampled using a Gaussian distribution. For Gaussian Naive Bayes the probability is calculated as follows:

$$P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right) \quad (2)$$

- 3) **Logistic Regression** It is a supervised machine learning model widely used for binary classification. The main

aim of this algorithm is to come up with a decision boundary that separates the data-points or instances into either of the two classes. The Logistic Response Function is the hypothesis used in this algorithm which converts the decision boundary into the probability of belonging to a particular class. The hypothesis or the logistic response function is as follows:

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T X}} \quad (3)$$

where

$$\theta^T X = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n \quad (4)$$

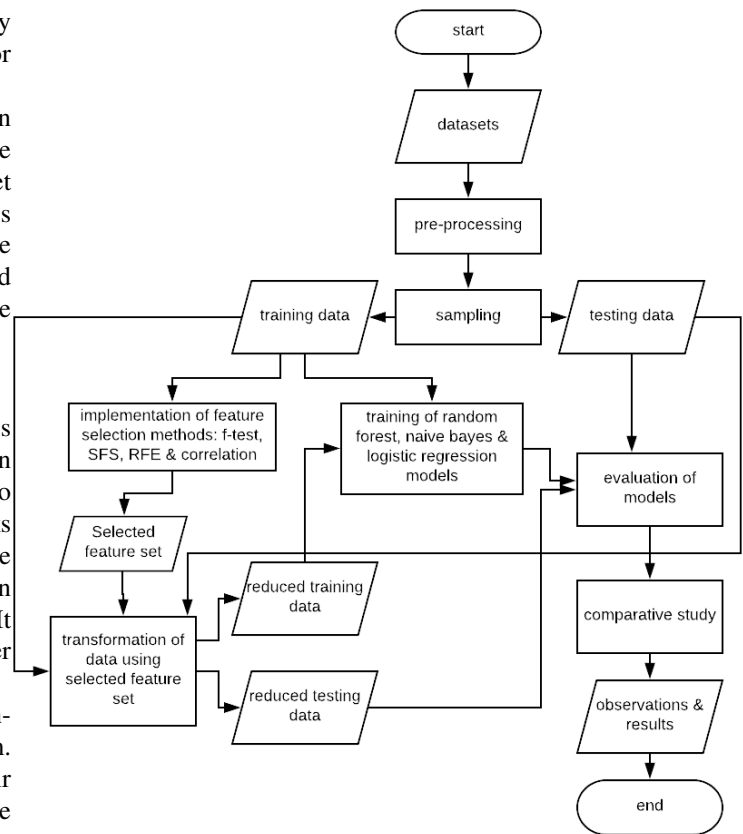


Fig. 1. Flowchart for proposed method

IV. EXPERIMENTAL RESULTS

A. Dataset without feature selection

We have applied the machine learning algorithms on the 2 datasets without using feature selection algorithms. Table1, Figure 2 and Table 2, Figure 3 show the classification accuracies obtained for Wisconsin and WDBC datasets respectively. We obtained the maximum accuracy for Random Forest algorithm and the least accuracy for Naive Bayes in both the datasets. The reason for this is because random forest is an

ensemble algorithm which combines the results from many decision trees for better performance. Also the disadvantage of Naive Bayes is that it considers the features to be independent which may not be applicable in real-life data, whereas logistic regression tries to fit a line which separates the 2 classes which may fail in cases where the data is non-linear.

TABLE I
ACCURACY COMPARISON FOR BREAST CANCER WISCONSIN DATASET

Model	Accuracy
Random Forest	97.142857
Naive Bayes	95.714286
Logistic Regression	96.428571

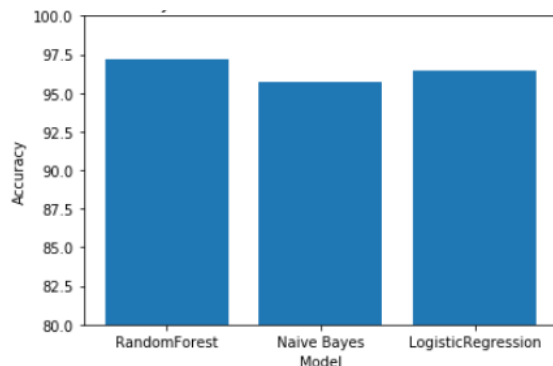


Fig. 2. Accuracy for dataset 1.

TABLE II
ACCURACY COMPARISON FOR WDBC DATASET

Model	Accuracy
Random Forest	97.368421
Naive Bayes	92.982456
Logistic Regression	95.614035

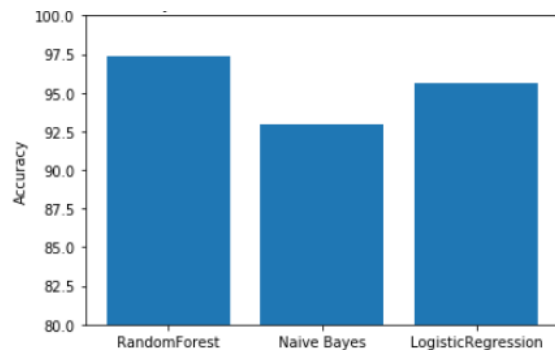


Fig. 3. Accuracy for dataset 2.

B. Dataset with feature selection

Breast cancer classification has been performed with the following four feature selection techniques- RFE, SFS, f-test, correlation - and the following classifiers- Random Forest, Naive Bayes and Logistic Regression. The result have been shown in Table 3,4,5, Fig 4,5,6 and Table 6,7,8, Fig 7,8,9 for Wisconsin and WDBC datasets respectively.

For each combination of feature selection algorithm and classifier a dataset size vs accuracy graph has been plotted and the maximum accuracy in each case has been considered for the comparative analysis. In case of correlation feature selection technique the correlation threshold has been set as 0.9.

TABLE III
ACCURACIES GIVEN BY RANDOM FOREST FOR DIFFERENT FEATURE SELECTION ON WISCONSIN

Feature Selection Algorithm	Max accuracy	No.of features
f-test	97.14	5,7,8,9
RFE	97.14	5,7,8,9
SFS	-	-
Correlation	97.85	8

TABLE IV
ACCURACIES GIVEN BY NAIVE BAYES FOR DIFFERENT FEATURE SELECTION ON WISCONSIN

Feature Selection Algorithm	Max accuracy	No.of features
f-test	97.14	5,6,7,8
RFE	-	-
SFS	95.71	2,8,9
Correlation	95.71	8

TABLE V
ACCURACIES GIVEN BY LOGISTIC REGRESSION FOR DIFFERENT FEATURE SELECTION ON WISCONSIN

Feature Selection Algorithm	Max accuracy	No.of features
f-test	97.14	6
RFE	97.14	5
SFS	97.14	4,5,6,7
Correlation	96.42	8

TABLE VI
ACCURACIES GIVEN BY RANDOM FOREST FOR DIFFERENT FEATURE SELECTION ON WDBC

Feature Selection Algorithm	Max accuracy	No.of features
f-test	97.36	16,19,22,28,30
RFE	98.24	16
SFS	-	-
Correlation	94.73	20

The reduced feature set which yields the maximum accuracy for each classifier using various feature selection algorithms mentioned above are different as shown in tables 3, 4, 5 for Wisconsin dataset and tables 6, 7, 8 for WDBC dataset. It proves that the entire feature set is not required and in many

TABLE VII
ACCURACIES GIVEN BY NAIVE BAYES FOR DIFFERENT FEATURE
SELECTION ON WDBC

Feature Selection Algorithm	Max accuracy	No.of features
f-test	96.49	10
RFE	-	-
SFS	98.24	3
Correlation	89.47	20

TABLE VIII
ACCURACIES GIVEN BY LOGISTIC REGRESSION FOR DIFFERENT FEATURE
SELECTION ON WDBC

Feature Selection Algorithm	Max accuracy	No.of features
f-test	95.61	16-30
RFE	95.61	23-30
SFS	96.49	4
Correlation	91.22	20

cases almost half the feature set gives better accuracy than the entire dataset. From tables 3 and 4 it is seen that Naive Bayes does not work with RFE since Naive Bayes does not give feature importance which is necessary for the working of RFE and though Random Forest works with SFS it is not computationally feasible due to high time complexity.

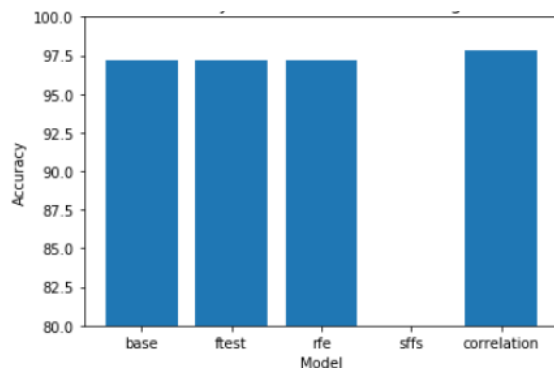


Fig. 4. Random Forest for Breast cancer Wisconsin

From fig 4, it is clear that after applying correlation feature selection algorithm, the accuracy of random forest classification algorithm shifts up from 97.14 to 97.85. This is because correlation algorithm removes the redundant features thereby improving the performance of the classifier.

Fig 5 shows that the combination of Naive Bayes Classifier with f-test feature selection algorithm increases the accuracy from 95.7142 to 96.49 percent on Wisconsin data set which is the maximum accuracy obtained for Naive Bayes on Wisconsin dataset.

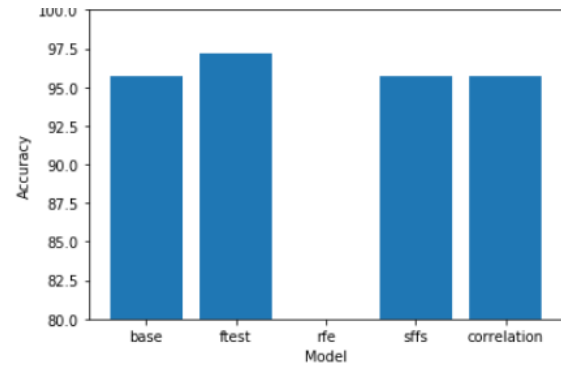


Fig. 5. Naive Bayes for Breast cancer Wisconsin

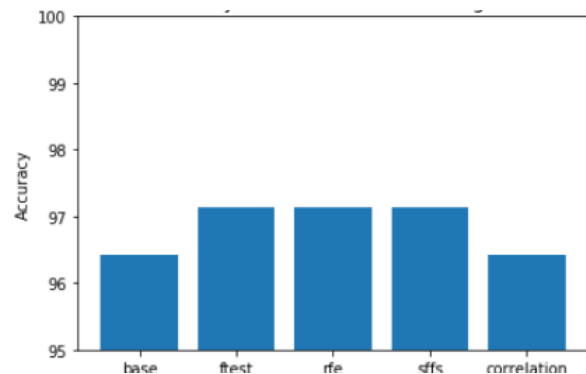


Fig. 6. Logistic Regression for Breast cancer Wisconsin

Fig 6 shows that the combination of Logistic Regression Classifier with f-test, rfe and sfs feature selection algorithm increases the accuracy from 95.6140 to 97.14 percent on Wisconsin data set whereas correlation yields the same accuracy as that of the base classifier.

From fig 7 we can see that the combination of Random Forest Classifier with rfe feature selection algorithm increases the accuracy from 97.36 to 98.24 percent on Wisconsin data set which is the maximum accuracy obtained for Naive Bayes on WDBC dataset.

From fig 8 we can see that the combination of Naive Bayes Classifier with sfs feature selection algorithm increases the accuracy from 92.98 to 98.24 percent on Wisconsin data set which is the maximum accuracy obtained for Naive Bayes on WDBC dataset.

From fig 9 we can see that the combination of Logistic Regression Classifier with sfs feature selection algorithm increases the accuracy from 95.61 to 97.14 percent on Wisconsin data set which is the maximum accuracy obtained for Naive Bayes on WDBC dataset.

From fig 7, 8, 9 it can be seen that for WDBC dataset the accuracies of the base classifiers after applying correlation feature selection algorithm is consistently less compared to those obtained using the base classifiers alone. This might be because correlation feature selection algorithm could have eliminated features that are relevant for the prediction.

Hence it is evident that Random Forest with feature selec-

tion algorithms stands for predicting more accurate results. When all the feature selection algorithms are taken into consideration, f-test feature selection is consistently successful at improving the accuracy of base classifiers for Breast cancer Wisconsin dataset and SFS feature selection is consistently successful at improving the accuracy of base classifiers for WDBC dataset. Table 9 and Table 10 gives the accuracies for different machine learning classifiers using feature selection for Wisconsin and WDBC datasets respectively. So, feature selection algorithms prove to influence the accuracy of the classification algorithm used for prediction.

TABLE IX
ACCURACY COMPARISON FOR BREAST CANCER WISCONSIN DATASET
WITH FEATURE SELECTION

	f-test	RFE	SFS	Correlation
Random Forest	97.14	97.14	-	97.85
Naive Bayes	97.14	-	95.71	95.71
Logistic Regression	97.14	97.14	97.14	96.42

TABLE X
ACCURACY COMPARISON FOR WDBC DATASET WITH FEATURE
SELECTION

	f-test	RFE	SFS	Correlation
Random Forest	97.36	98.24	-	94.73
Naive Bayes	96.49	-	98.24	89.47
Logistic Regression	95.61	95.61	96.49	91.22

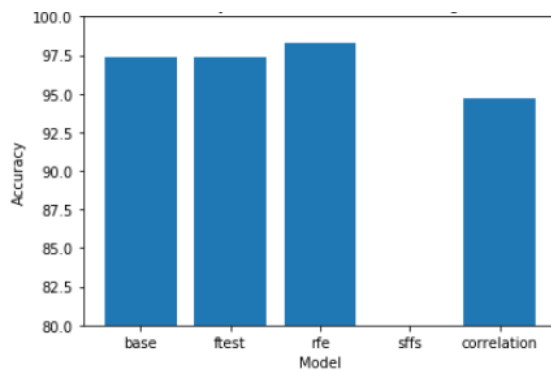


Fig. 7. Random Forest for WDBC

V. CONCLUSION

In this study we can see that we obtain the same accuracy for a subset of the entire feature set. From this we can conclude that not all the features are necessary for breast cancer prediction and feature selection helps in building an efficient model in such scenarios. Though feature selection is effective in general, it will not improve the performance of the

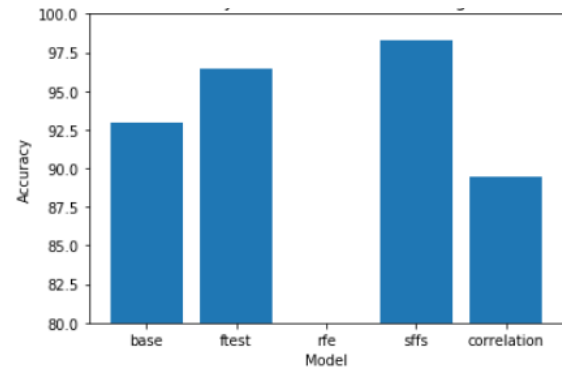


Fig. 8. Naive Bayes for WDBC

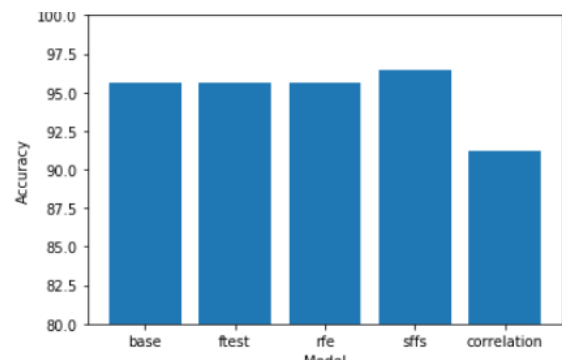


Fig. 9. Logistic Regression for WDBC

model if the feature selection algorithm excludes features relevant for classification. In conclusion though Random Forest classifier gives the highest accuracy with feature selection for both the datasets, f-test which is a filter method consistently improves the accuracy of the base classifiers as compared to the other feature selection techniques for Breast cancer Wisconsin dataset whereas SFS which is a wrapper method consistently improves the accuracy of the base classifiers for WDBC dataset.

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